FOODWISE AI

AI-Powered Food Waste Prediction & Redistribution Platform

Mini Project Documentation

Tagline
Predict Better. Waste Less. Feed More.

A web-based AI application designed to help restaurants, college canteens, hostels, hotels, cafeterias, and food-service organizations reduce food waste through intelligent demand prediction, waste-risk analysis, surplus detection, and food redistribution.

Project Type: Mini Project
Domain: Artificial Intelligence & Machine Learning
Application Type: Web Application
Development Platform: Base44 AI
Live Application: https://cuddly-food-wise-flow.base44.app

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1. Project Overview

Food waste is a major social, economic, and environmental problem. Restaurants, college canteens, hostels, hotels, and event organizers often prepare more food than is actually consumed because future demand is difficult to estimate.

At the same time, edible surplus food may still have value for people and organizations that need it.

FoodWise AI addresses this problem by providing an intelligent platform that analyzes historical consumption patterns and operational data to estimate future food demand and identify potential food surplus.

```
Historical Data
↓
AI Demand Prediction
↓
Waste Risk Analysis
↓
Production Recommendation
↓
Surplus Detection
↓
Redistribution Recommendation
↓
Impact Measurement
↓
Continuous Improvement
```

2. Problem Statement

Food-service organizations frequently face difficulties in accurately estimating the quantity of food required for a particular day or meal.

Overestimating demand results in:

• Excess food preparation

• Food wastage

• Financial losses

• Unnecessary resource consumption

• Increased environmental impact

Underestimating demand can result in:

• Food shortages

• Customer dissatisfaction

• Emergency preparation

• Increased operational costs

There is a need for an intelligent system that can analyze historical consumption and operational patterns to provide data-driven food demand predictions and waste-reduction recommendations.

3. Proposed Solution

FoodWise AI provides a centralized web platform where organizations can record food preparation, consumption, and waste information.

The system analyzes this data and provides:

• Food demand predictions

• Recommended production quantities

• Waste-risk scores

• Surplus detection

• AI-generated insights

• Food redistribution recommendations

• Impact analytics

• Sustainability reports

Instead of simply recording food waste after it occurs, FoodWise AI focuses on predicting and preventing waste before it happens.

4. Objectives

• Predict food demand using historical consumption and operational data.

• Reduce food waste by recommending appropriate production quantities.

• Identify potential surplus food before it becomes unnecessary waste.

• Recommend redistribution opportunities for usable surplus food.

• Provide explainable AI insights so users understand why a prediction or recommendation was generated.

• Track environmental and financial impact through analytics and reports.

• Continuously improve predictions by comparing predicted demand with actual consumption.

5. Key Features

AI Demand Forecasting

Analyzes historical food-consumption patterns and predicts expected demand.

Waste Risk Prediction

Calculates the probability of excessive food waste and presents risk levels and contributing factors.

Production Recommendation

Recommends how much food should be prepared based on predicted demand and safety requirements.

Waste Monitoring

Tracks food prepared, consumed, remaining, and discarded, with daily, weekly, and monthly analytics.

Surplus Food Detection

Identifies potential surplus when prepared food exceeds consumption.

Food Redistribution

Allows food providers to publish surplus food and enables registered NGOs or redistribution organizations to request it.

AI Redistribution Recommendation

Recommends suitable redistribution partners based on factors such as distance, quantity, food type, availability, and pickup deadline.

Impact Analytics

Tracks meals saved, food waste reduced, meals redistributed, estimated savings, and environmental impact.

Explainable AI

Provides understandable reasons behind predictions and recommendations.

AI Insights

Generates insights from historical activity to support waste-reduction decisions.

6. AI/ML Components

6.1 Demand Forecasting

Predicts future food demand based on historical information such as consumption, customer count, day of week, meal type, seasonal trends, events, and previous surplus.

6.2 Waste Risk Prediction

Estimates the probability that prepared food will become waste by considering planned production, predicted demand, historical waste, and consumption trends.

6.3 Recommendation System

Generates recommendations for production quantity, waste reduction, surplus handling, and redistribution partners.

6.4 Anomaly Detection

Identifies unusual changes in consumption or waste, such as a sudden increase in discarded food.

6.5 Explainable AI

Provides understandable factors behind AI predictions and recommendations, helping users trust and interpret system outputs.

7. System Workflow

```
Food Provider
↓
Historical Food Data
↓
AI Analysis
```

```
↓
Demand Forecast + Waste Risk + Consumption Analysis
↓
Production Recommendation
↓
Food Consumption
↓
Surplus Detection
↓
Redistribution Recommendation
↓
NGO / Partner
↓
Food Redistribution
↓
Impact Analytics
↓
AI Learning
```

8. Application Modules

Module 1 — Authentication

Provides user access through sign up, login, user accounts, and role-based access.

Module 2 — Organization Management

Maintains organization information, food-service information, operating details, and sustainability goals.

Module 3 — Food Data Management

Records food items, quantity prepared, quantity consumed, quantity remaining, and quantity wasted.

Module 4 — AI Demand Forecast

Generates predicted demand, recommended production, expected surplus, and confidence score.

Module 5 — Waste Monitoring

Provides waste trends, food category analysis, waste percentage, and historical comparisons.

Module 6 — Surplus Management

Detects surplus, creates surplus listings, sets pickup deadlines, and tracks surplus status.

Module 7 — Redistribution

Connects food providers with NGOs or redistribution partners and tracks the donation workflow.

Module 8 — Analytics

Provides visual analytics for consumption, waste, demand, predictions, savings, redistribution, and impact.

Module 9 — AI Insights

Generates personalized recommendations based on organization data.

Module 10 — Reports

Provides daily, weekly, monthly, redistribution, and sustainability reports.

9. User Roles

Food Provider

• Record food production

• Track consumption

• Monitor waste

• View AI predictions

• Create surplus listings

• Accept redistribution requests

• View impact analytics

NGO / Redistribution Partner

• View available surplus

• Request food

• Schedule pickups

• Track donations

• View redistribution history

Administrator

• Manage users

• Manage organizations

• Manage NGOs

• Monitor redistribution

• View system-wide analytics

• Manage reports

10. Technology Stack

Layer	Technology / Approach
Development Platform	Base44 AI
Frontend	Responsive web interface and dashboard components
Backend	Base44 managed application logic and backend capabilities
Database	Base44 data/entity storage
AI	AI-powered demand analysis, prediction, recommendation, and explainable insights

Application Type	Responsive Web Application

11. Application Architecture

```
USER INTERFACE
Dashboard | Forecast | Waste | Surplus | Analytics | Reports
↓
APPLICATION LOGIC
Authentication | Data Processing | Forecasting | Recommendations
↓
AI LAYER
Demand Prediction | Waste Risk | Recommendation | Anomaly Detection
↓
DATA / BACKEND
Users | Organizations | Food Records | Consumption | Waste
Predictions | Surplus | Redistribution
```

12. Database Design

Users

User ID, Name, Email, Authentication Reference, Role, Organization ID

Organizations

Organization ID, Organization Name, Organization Type, Location, Capacity, Average Customers

Food Records

Food ID, Organization ID, Food Name, Category, Quantity Prepared, Quantity Consumed, Quantity Remaining, Quantity Wasted, Date

Predictions

Prediction ID, Organization ID, Date, Predicted Demand, Recommended Production, Confidence, Waste Risk

Surplus

Surplus ID, Organization ID, Food Type, Quantity, Pickup Location, Pickup Deadline, Status

Redistribution

Request ID, Surplus ID, NGO ID, Requested Quantity, Request Status, Pickup Status, Completion Date

13. AI Prediction Workflow

```
Historical Data
↓
Data Analysis
↓
Pattern Detection
↓
Demand Forecast
↓
Waste Risk Calculation
↓
Production Recommendation
```

```
↓
Actual Consumption
↓
Prediction vs Actual
↓
Error Analysis
↓
Future Improvement
```

14. Example AI Scenario

Consider a college canteen with the following historical meal demand:

```
Monday → 410 meals
Tuesday → 425 meals
Wednesday → 420 meals
Thursday → 430 meals
Friday → 440 meals
```

The AI analyzes the trend and predicts:

```
Expected Demand: 438 meals
Recommended Production: 450 meals
Expected Surplus: 12 meals
Waste Risk: LOW — 18%
AI Confidence: 91%
```

Example explanation:

- Historical demand is stable.
- Similar days averaged approximately 425–440 meals.
- Recent customer traffic increased.
- A small safety buffer is included.

15. Expected Benefits

Environmental Benefits

- Reduced food waste
- Reduced unnecessary resource consumption
- Better utilization of prepared food
- Increased awareness of sustainability

Economic Benefits

- Reduced food-production costs
- Reduced disposal costs
- Better resource planning
- Potential cost savings

Social Benefits

• Increased redistribution of usable surplus food

• Improved coordination between providers and NGOs

• Better utilization of available food

16. Future Scope

• Real-time demand prediction using live customer or order data.

• Weather integration to improve demand prediction.

• Event-based forecasting for holidays, festivals, college events, and conferences.

• Computer vision to estimate leftover food quantities.

• IoT integration with smart weighing systems.

• Mobile applications for providers, NGOs, and pickup personnel.

• Route optimization for redistribution partners.

• Advanced ML models such as Random Forest, XGBoost, LSTM, and other time-series forecasting approaches.

• Real-time notifications when suitable surplus food becomes available.

• A larger multi-organization network connecting restaurants, canteens, NGOs, and communities.

17. Limitations

• Prediction quality depends on the quality and quantity of historical data.

• Demonstration datasets may not represent real-world food-demand behavior.

• Real-world redistribution requires food-safety compliance and operational coordination.

• Environmental impact calculations should use validated region-specific methodologies before being treated as official measurements.

• Production-level forecasting would require larger and more diverse datasets.

18. Academic Relevance

FoodWise AI demonstrates concepts relevant to B.Tech CSE (AI/ML), including:

• Artificial Intelligence

• Machine Learning

• Predictive Analytics

• Time-Series Forecasting

• Recommendation Systems

• Data Analytics

• Explainable AI

• Database Management

• Web Application Development

• Role-Based Access

• Sustainable Technology

• Human-Centered AI

19. Project USP

The major uniqueness of FoodWise AI is that it does not simply record food waste after it happens. Instead, it aims to:

```
PREDICT → PREVENT → REDISTRIBUTE → MEASURE → IMPROVE
```

Core idea:

“Don't wait for food waste to happen. Predict it before it happens.”

20. Conclusion

FoodWise AI presents an intelligent approach to reducing food waste by combining AI-powered prediction, data analytics, recommendation systems, and surplus-food redistribution.

Instead of treating food waste as a problem only after it occurs, the platform focuses on early prediction and preventive decision-making.

By analyzing historical consumption patterns and operational data, FoodWise AI can help organizations make better production decisions, identify potential surplus, and connect usable surplus food with redistribution partners.

The project demonstrates how Artificial Intelligence can be applied to a real-world sustainability and social-impact problem.

FoodWise AI — Predict Better. Waste Less. Feed More.

21. References

1. Base44 AI — AI-powered application development platform.

2. FoodWise AI Live Application — https://cuddly-food-wise-flow.base44.app

3. General concepts of demand forecasting, predictive analytics, recommendation systems, and food-waste reduction.